AP Physics 1 - Unit 3 - Lesson 1

What is energy? Types of energy

STUDENT WORK-PACK

Name: _____ Date: _____

How to use this pack
Work through the matching HTML file. Write your thinking here before opening a hint. Use the worked solution only after you have made a serious attempt. Circle or mark anything you want to ask about at the end.

Unit 3 road map

L1	What is energy? Types of energy
L2	Energy transfers and conservation of energy
L3	Power of an energy-transfer machine
L4	Mechanical work
L5	Spring-energy derivation
L6	Work-energy theorem and systems
L7	Applications: inclines, Atwood systems, etc.
L8	Assessed lab
L9	Review
L10	UNIT TEST - 16/17 November 2026

Proposed unit test: 16/17 November 2026

Today's objectives

• Describe energy as a conserved scalar quantity used to account for the state and changes of a physical system.

• Recall the SI unit of energy and the GREENSICK memory aid for common forms of energy.

• Distinguish broadly between kinetic and potential energy and explain mechanical energy.

• Match the AP-style energy symbols to their meanings and select/use the correct energy equation.

• Use SI units, convert common non-SI units, and rearrange energy equations.

1. What is energy?

Before revealing the HTML note: complete the sentence in your own words.

Energy is ...

Its SI unit is _____ and the unit symbol is _____.

A useful model answer after discussion:

Energy is a _____ quantity used to account for the state of an object/system and the changes that can occur. Energy can be _____ or _____, but the total energy is conserved.

2. GREENSICK - nine familiar forms/stores

Complete the mnemonic. The first letter is provided. This is a memory aid rather than a formal College Board classification.

Letter	Energy form/store	Kinetic / potential / other description?
G	_____	_____
R	_____	_____
E	_____	_____
E	_____	_____
N	_____	_____
S	_____	_____
I	_____	_____
C	_____	_____
K	_____	_____

Big idea: kinetic energy is associated with motion. Potential energy belongs to a system and depends on the relative positions/configuration of interacting objects. Many everyday “forms” of energy are deeper combinations of microscopic kinetic and potential energies.

3. AP-style symbols - match them

Choose from this symbol bank:

K	K_R	ΔU_g	U_g	U_s	ΔU_{int}	E_{mech}

Meaning	Symbol
translational kinetic energy	_____
rotational kinetic energy	_____
change in gravitational potential energy near a planet surface	_____
gravitational potential energy for two spherical masses	_____
spring (elastic) potential energy	_____
change in internal energy (bookkeeping notation used in this course)	_____
mechanical energy = kinetic + potential energy	_____

4. Build the energy equation toolkit

Translational kinetic energy

$$K = \tfrac{1}{2}mv^2$$

SI inputs: m in kg; v in m s^{-1}

Meaning / condition for use: _____

Rotational kinetic energy

$$K_R = \tfrac{1}{2}I\omega^2$$

SI inputs: I in kg m^2; ω in rad s^{-1}

Meaning / condition for use: _____

Near-surface gravitational PE change

$$\Delta U_g = mg\Delta y$$

SI inputs: m in kg; g in N kg^{-1}; Δy in m

Meaning / condition for use: _____

General gravitational PE

$$U_g = -Gm_1m_2/r$$

SI inputs: G = 6.67 × 10^{-11} N m^2 kg^{-2}; r in m

Meaning / condition for use: _____

Spring potential energy

$$U_s = \tfrac{1}{2}k(\Delta x)^2$$

SI inputs: k in N m^{-1}; Δx in m

Meaning / condition for use: _____

Internal-energy change

$$\Delta U_{int} = U_{int,f} - U_{int,i}$$

SI inputs: all energy values in J

Meaning / condition for use: _____

Mechanical energy

$$E_{mech} = K + U$$

SI inputs: all terms in J

Meaning / condition for use: _____

Mechanical energy: What kinds of energy are included?

_____

Not today: we are deliberately not using the mechanical-work equation yet. Mechanical work is Lesson 4.

5. Practice Round A - direct substitution (all quantities already in SI)

For each: (1) identify the energy type/equation, (2) substitute with units, (3) calculate, (4) give the final answer in joules.

A1. Translational kinetic energy
A 2.0 kg cart moves at 3.0 m s^{-1}. Determine its translational kinetic energy.

A2. Rotational kinetic energy
A wheel has rotational inertia I = 0.50 kg m^2 and angular speed ω = 4.0 rad s^{-1}. Determine its rotational kinetic energy.

A3. Gravitational PE change
A 3.0 kg object is raised vertically by 2.0 m near Earth. Use g = 9.8 N kg^{-1}. Determine ΔU_g.

A4. General gravitational PE
A 1000 kg satellite is 7.00 × 10^6 m from Earth's center. Take Earth's mass as 5.97 × 10^{24} kg. Determine U_g.

A5. Spring potential energy
An ideal spring has k = 200 N m^{-1} and is compressed by 0.10 m. Determine U_s.

A6. Internal-energy change
A system’s internal energy changes from 250 J to 310 J. Determine ΔU_{int}.

A7. Mechanical energy
A system has K = 20 J, U_g = 35 J, and U_s = 5 J. Determine E_{mech}.

6. Practice Round B - convert to SI first

Write each conversion explicitly before you substitute into the energy equation.

B1. Translational kinetic energy
A 750 g ball moves at 18 km h^{-1}. Determine K in joules.

B2. Rotational kinetic energy
A rotor has I = 0.020 kg m^2 and rotates at 120 rpm. Determine K_R.

B3. Gravitational PE change
A 650 g mass is lifted through 45 cm. Use g = 9.8 N kg^{-1}. Determine ΔU_g.

B4. General gravitational PE
A 2.0 tonne satellite is 7000 km from Earth's center. Use Earth mass 5.97×10^{24} kg. Determine U_g.

B5. Spring potential energy
A spring has $k = 3.0\ \text{N cm}^{-1}$ and is compressed by 6.0 cm. Determine U_s.

B6. Internal-energy change
A system's internal energy changes from 0.45 kJ to 620 J. Determine ΔU_{int} in joules.

B7. Mechanical energy
A system has K = 0.80 kJ, $U_g = 350$ J, and $U_s = 0.050$ kJ. Determine E_{mech} in joules.

7. Practice Round C - rearrange to find a missing quantity

Rearrange symbolically first whenever possible, then substitute.

C1. Translational kinetic energy
An 8.0 kg object has K = 64 J. Determine its speed.

C2. Rotational kinetic energy
A rotor has K_R = 9.0 J and I = 0.50 kg m². Determine ω.

C3. Gravitational PE change
A 5.0 kg object gains 196 J of gravitational potential energy near Earth. Use g = 9.8 N kg⁻¹. Determine Δy.

C4. General gravitational PE
A 1.00 × 10³ kg spacecraft has U_g = −6.67 × 10⁹ J in the field of a planet of mass 5.00 × 10²⁴ kg. Determine r.

C5. Spring potential energy
A spring with $k = 100\ \text{N m}^{-1}$ stores 4.5 J. Determine $|\Delta x|$.

C6. Internal-energy change
A system starts with $U_{int,i} = 120$ J and has $\Delta U_{int} = +45$ J. Determine $U_{int,f}$.

C7. Mechanical energy
A system has $E_{mech} = 90$ J, $K = 35$ J, and $U_g = 40$ J. The only other mechanical-energy store is a spring. Determine U_s.

8. End-of-lesson check

☐ I can explain the difference between kinetic and potential energy.

☐ I can identify mechanical energy as the sum of kinetic and potential energies in the chosen system.

☐ I can match K, K_R, ΔU_g, U_g, U_s, ΔU_{int} and E_{mech} to their meanings.

☐ I can choose an appropriate energy equation and use SI units.

☐ I can convert non-SI quantities before calculating energy.

☐ I can rearrange the energy equations to determine an unknown quantity.

Questions I still have

AP alignment note: The current AP Physics 1 framework explicitly includes translational kinetic energy, spring potential energy, near-surface and general gravitational potential energy, conservation/mechanical energy, and power. Rotational kinetic energy is formally developed later in Unit 6. ΔU_{int} is used here as bookkeeping notation for internal-energy change; the current AP Physics 1 CED expects awareness that mechanical energy can be dissipated as thermal energy or sound.